

Design Spec: Quick Desktop AI Control Tower

Executive Summary

This design spec defines the end-to-end experience for the Quick Desktop AI Control Tower — a cross-vendor AI governance capability embedded within Amazon Quick Desktop. The design targets enterprise IT operations leaders (primary: Lisa Nakamura, Director of Digital Workplace) who need to discover shadow AI, enforce governance policies, attribute costs, and measure adoption maturity from a single desktop-native console. The experience follows Salesforce+Datadog enterprise patterns (dense information surfaces, contextual data surfacing, lateral navigation) implemented through Cloudscape Design System components. The design covers seven navigable sections spanning five governance dimensions (Discover, Observe, Govern, Secure, Measure) plus Admin Console and Integration Marketplace, meeting o

Design Context

Type: Enterprise

Patterns Applied: Salesforce+Datadog Hybrid

Component Library: Cloudscape Design System

Primary Persona: Lisa Nakamura, Director of Digital Workplace (medium-high technical sophistication, manages 8 IT ops staff, not writing CloudWatch queries)

Secondary Persona: David Park, VP Engineering (high technical sophistication, former SRE)

Phase 1: First Principles Design

Primary User Task

Gain complete visibility into organizational AI tool usage — what AI exists, what it costs, whether it complies with policy, and whether the investment is delivering value — and take governance action directly from the desktop agent employees already use daily.

Information Priority

Governance health at a glance — aggregate KPIs (total AI tools discovered, % governed, total spend, compliance rate, maturity score) displayed as a hero bar the moment the user enters the Control Tower. This answers the CEO's question in 5 seconds.

Shadow AI alerts — newly discovered unapproved tools and active security risk notifications, surfaced as a priority attention zone below the hero bar. This is the "aha" moment that drives adoption.

AI inventory drill-down — the complete catalog of AI tools with per-tool detail (who uses it, cost, compliance status, risk level) accessible via table with split-panel detail. This is the working surface for daily governance.

Trend and cost data — adoption trends over time, cost attribution by team/vendor, maturity progression — the reporting layer that supports quarterly reviews and executive presentations.

Policy and configuration — governance policy management, integration setup, and admin settings — accessed less frequently but critical for initial setup and ongoing tuning.

Critical User Journey

Step	User Action	System Response	Data Shown
1	Opens Quick Desktop, navigates to AI Control Tower via sidebar	Control Tower Command Center loads with pre-aggregated data	Hero KPI bar: 247 AI tools, \$182K/mo spend, 72% governed, 3 active risks, maturity level 3
2	Scans the shadow AI alert zone	System highlights 4 newly discovered unapproved tools from the last 7 days	Tool names, team using them, data sensitivity risk, recommended action
3	Clicks an unapproved tool (e.g., "Jasper AI — Marketing")	Split panel opens with tool detail	Tool profile: 12 users, \$2.4K/mo estimated cost, no governance policy, medium data risk, discovery source (desktop detection)
4	Clicks "Apply Policy" on the tool detail	Policy selection modal appears with recommended governance action	Options: Approve (add to sanctioned list), Monitor (track but don't block), Block (add to restricted list + notify users)
5	Selects "Approve with conditions" and configures data sensitivity nudge	System creates policy, schedules user notifications	Confirmation: policy applied, 12 affected users will see compliant-usage nudge within 24 hours
6	Navigates to Cost Attribution tab	Cost dashboard loads with cross-vendor breakdown	Stacked bar by vendor (AWS Bedrock \$48K, Copilot \$72K, OpenAI \$38K, Others \$24K), table by team with per-user drill-down
7	Returns to Command Center, verifies updated governance %	KPI hero bar refreshes (73% governed, 3→2 active risks)	Updated metrics reflecting the policy action just taken

Decision Points

Decision	Data Required	Design Element
Should this AI tool be approved or blocked?	Tool name, vendor, user count, data sensitivity assessment, cost estimate, comparable approved alternatives	Split-panel tool detail with risk assessment card + "Apply Policy" action button
Which team is overspending on AI?	Cost-by-team breakdown, vendor attribution, per-user cost, trend vs. prior period	Cost Attribution dashboard with horizontal bar chart by team, drill-down table
Is the organization improving its governance posture?	Maturity score trend, % governed trend, compliance rate over time, number of open risks	Command Center sparklines in hero bar + Maturity tab with 5-level scorecard
Should we consolidate redundant AI tools?	Tools with overlapping capabilities, user overlap between tools, cost comparison	AI Inventory with "Duplicate Detection" badge and comparison view in split panel
Are governance nudges effective or annoying?	Override rate, nudge acceptance rate, user feedback, compliance trend post-nudge	Governance tab metrics section with override-rate anti-metric prominently displayed

Minimum Viable Interaction

2 clicks from landing to the "aha" moment:

Click "AI Control Tower" in Quick Desktop sidebar navigation → Command Center loads with all KPIs and shadow AI alerts
Click any shadow AI alert → full tool detail with governance action options

3 clicks from landing to governance action:

Click "Apply Policy" → policy applied, users notified

Phase 2: Pattern Reality Check

Applied Patterns

Salesforce — Contextual Data Surfacing: The AI Inventory page follows the Salesforce record-page pattern. Clicking any AI tool opens a split-panel detail view that proactively surfaces related data: users of the tool, cost data, compliance status,

similar tools, governance history, and recommended actions. The user never needs to navigate away to find context about a tool — everything related to it appears in the detail panel.

Datadog — Information Density: The Command Center hero bar packs five KPIs with sparkline trends into a single row, following Datadog's "maximum signal per pixel" principle. The AI Inventory table uses sparklines-in-table-cells for usage trends (per Datadog's enterprise density rule), allowing Lisa to scan adoption patterns across 50+ tools without opening individual detail views.

Datadog — Connectedness Rather Than Consistency: Every data point is a portal. Click a cost number → navigate to Cost Attribution filtered to that tool. Click a team name → see all tools that team uses. Click a risk alert → see the tool, the policy violation, and the affected users. Lateral, data-driven navigation throughout.

Salesforce — Configuration Over Customization: Saved views for the AI Inventory (Lisa's "Unapproved Tools" view, David's "Engineering AI Stack" view). Collection Preferences on every table for column visibility and density. Dashboard layout is opinionated by default but configurable per user.

Layout Rationale

Command Center as hub, not just a dashboard: The Command Center exists because Lisa's #1 JTBD is answering the CEO's AI question in under 5 minutes. A hub pattern (KPI hero bar → alert zone → dimension cards linking to sections) is chosen over a blank dashboard builder because (a) Lisa is medium-high technical but not building custom dashboards, (b) the PRD explicitly states "pre-configured and opinionated," and (c) ServiceNow's Command Center pattern is table stakes. The hub earns its place by being the single screen that answers "what's the state of our AI?"

Split-panel detail over full-page navigation: The AI Inventory uses a split panel (Cloudscape Split Panel component) rather than navigating to a separate detail page because (a) enterprise users need to compare tools by switching between rows quickly, (b) the Datadog "connectedness" principle says context should never be lost, and (c) Cloudscape's Split Panel is the standard pattern for list-detail in AWS consoles.

Sidebar navigation over top nav: Cloudscape App Layout uses sidebar navigation, and all AWS console products follow this pattern. Additionally, the seven-section navigation is too deep for horizontal tabs but fits naturally in a collapsible sidebar.

Hero KPI bar over separate metrics page: The five governance dimensions each have a KPI that matters most. Showing all five in a hero bar above the fold means Lisa never needs to navigate to a separate metrics page for the executive summary. This follows the Datadog principle of density — the answer to "how are we doing" is visible in 0 clicks after entering the Control Tower.

Alternatives Considered

Decision	Chosen	Alternative A	Alternative B	Why Chosen Wins
Information architecture	Hub + section navigation (7 sections in sidebar)	Single-page dashboard with tabs	Wizard-style guided flow	Hub + sections provides both at-a-glance executive summary AND deep-dive capability. Single-page dashboard can't accommodate 7 sections without tab overload. Wizard forces linear flow when Lisa often
Tool detail pattern	Split panel (side position)	Full-page tool detail	Modal overlay	Split panel preserves list context (see other tools while viewing detail). Full-page loses context. Modal blocks interaction with the underlying page.
Command Center layout	Pre-configured opinionated KPI hub	Blank dashboard builder (Grafana-style)	Card grid with drag-and-drop	Lisa doesn't build dashboards — she answers CEO questions. Opinionated defaults serve the 80% case; Cloudscape Collection Preferences handle the 20% customization need.

Decision	Chosen	Alternative A	Alternative B	Why Chosen Wins
Governance action UX	Inline "Apply Policy" in split panel detail	Separate policy management page	Bulk action toolbar on table	Inline action follows Salesforce's "act where you are" principle. Separate page adds 2+ clicks. Bulk action exists too (for power users), but the primary flow is individual tool governance from the de
Cost visualization	Stacked bar by vendor + table by team	Pie chart by vendor	Treemap by team/tool	Stacked bar handles 5+ vendors without the pie chart > 5 slices anti-pattern. Table provides the drill-down density enterprise users need. Treemap is visually interesting but harder to read precise va

Cloudscape Component Mapping

UI Element	Cloudscape Component	Notes
Page shell	App Layout	Side navigation, split panel, content area. Breadcrumbs in header.
Primary navigation	Side Navigation	7 items: Command Center, AI Inventory, Governance, Security, Cost Attribution, Adoption & Maturity, Admin Console. Expandable sections for Integration Marketplace, Compliance Reports, Enforcement.
KPI hero bar	Container + Box + Column Layout	5 stat cards in a row using Box with SpaceBetween. Each card: metric value, label, sparkline, trend indicator.
Alert zone	Flashbar	Dismissible alert items for shadow AI discoveries, policy violations, risk alerts. Type: warning/error/info.
AI tool table	Table + Collection Preferences + Property Filter	Sortable, filterable, paginated. Columns: tool name, vendor, users, monthly cost, compliance status, risk level, trend sparkline.
Tool detail	Split Panel	Side position on L/XL, bottom on M. Tabs inside: Overview, Users, Cost, Compliance, Activity Log.
Policy action	Modal (via Button trigger)	Form inside modal: policy type (approve/monitor/block), conditions, notification settings.
Cost charts	Container wrapping custom chart HTML	Stacked bar (by vendor), horizontal bar (by team). Chart.js or inline SVG with Cloudscape color tokens.
Maturity scorecard	Cards (in Grid layout)	5 maturity level cards, current level highlighted. Each card: level name, criteria checklist, % progress to next level.
Date range selector	Date Range Picker	Global time filter affecting all data views. Presets: Last 7 days, 30 days, 90 days, Custom.
Breadcrumbs	Breadcrumb Group	AI Control Tower > [Section] > [Subsection]
Page headers	Header	H1 with description, counter badge, action buttons
Governance policy table	Table + Collection Preferences	Rows: policy rules. Columns: rule name, scope (org/team/user), AI tools affected, enforcement level, status.
Integration cards	Cards (in Grid layout)	Per-connector cards: logo, name, status badge (connected/disconnected/error), user count, last sync.
Empty states	Box with illustration	Custom empty-state components per section with CTA and value prop.

UI Element	Cloudscape Component	Notes
Notification settings	Form (within Container)	Notification channel, frequency, severity threshold per alert type.
Tabs within sections	Tabs	Used inside Command Center (Overview, Trends), AI Inventory (All Tools, Shadow AI, Approved, Blocked).

Phase 3: Design Checklist + Audits

3A. Core Design Checklist

#	Criterion	Score	Evidence
1	Task Clarity	5	Primary task (answer CEO's AI question) achievable in 2 clicks: open Control Tower → read hero KPI bar. Governance action (approve/block tool) in 3 clicks.
2	Information Hierarchy	5	Hero KPI bar is the most visually prominent element (largest type, top of page). Shadow AI alerts are second (Flashbar with warning color). Drill-down table is third. Navigation is subordinate in side
3	Progressive Disclosure	4	Command Center shows summary → click section for detail → click row for split panel → click action for modal. Complexity hidden at each level. Deducted 1 point: advanced policy conditions might need b
4	Layout Rationale	5	Every layout decision documented with WHY in Phase 2 (hub pattern for CEO-question JTBD, split panel for context preservation, sidebar for 7-section depth, hero bar for zero-click executive summary).
5	Alternatives Considered	5	Five major layout decisions each evaluated against 2+ alternatives with rejection rationale (hub vs. blank dashboard, split panel vs. full page, pre-configured vs. builder, inline action vs. separate
6	Pattern Consistency	5	100% Cloudscape components. App Layout, Table, Split Panel, Property Filter, Flashbar, Modal, Cards, Tabs — all standard Cloudscape patterns. No custom components that deviate from the design system.
7	Stickiness	4	Morning briefing alert zone creates daily check-in habit. Shadow AI discovery provides ongoing "aha" moments. Maturity score gamification. Nudge override rate creates feedback loop. Deducted 1: could
8	Error States	4	Empty states designed per section (empty inventory, no integrations, no policies). Loading skeletons for data fetch. Error Flashbar for API failures. Timeout handling with retry. Deducted 1: partial d

#	Criterion	Score	Evidence
9	Expert vs. Novice	5	Novice: Command Center hub answers questions visually, 5-stage onboarding, empty states with CTAs. Expert: Property Filter with query syntax, keyboard shortcuts, saved views, collection preferences fo
10	Accessibility	4	WCAG AA target. Cloudscape components are pre-validated for contrast. ARIA landmarks defined. Keyboard nav spec included. Screen reader announcements defined. Deducted 1: charts need pattern/shape dif

Checklist Score: 10/10 pass (>= 3)

3B. Nielsen Heuristic Audit

#	Heuristic	Score	Evidence	Remediation
H1	Visibility of System Status	5	Hero KPI bar updates on every data refresh. Loading skeletons during fetch. Flashbar notifications for async operations (policy applied, scan complete). Date range picker shows active time window. Las	—
H2	Match Between System and Real World	4	Uses Lisa's language: "shadow AI" (not "unclassified assets"), "approved/blocked" (not "allow-listed/deny-listed"), "monthly cost" (not "token consumption"). Five governance dimensions match industry	Minor: ensure all vendor names use common brand names (ChatGPT not "OpenAI API"), maturity levels use plain English.
H3	User Control and Freedom	4	Policy actions are reversible (approve → block → approve). Split panel closes with X or Esc. Filters clear with "Clear All" button. Modal has Cancel. Governance nudge dismissal persists.	Minor: add "Undo" toast after policy changes with 10-second reversal window.
H4	Consistency and Standards	5	100% Cloudscape components ensure internal consistency. Table patterns identical across Inventory, Governance, Security sections. Split panel behavior identical everywhere. Follows AWS console convent	—
H5	Error Prevention	4	Destructive actions (block tool, delete policy) require confirmation modal with impact summary ("This will affect 12 users"). Governance nudge thresholds have sensible defaults. Integration setup vali	Minor: add "Are you sure?" for bulk policy actions affecting 50+ users.

#	Heuristic	Score	Evidence	Remediation
H6	Recognition Rather Than Recall	5	Property Filter shows token suggestions (tool names, vendors, teams). Saved views remember filter state. Breadcrumbs show current location. Side navigation highlights active section. Recently viewed t	—
H7	Flexibility and Efficiency of Use	4	Keyboard shortcuts (/ for search, Esc for close panel). Saved views for repeated queries. Collection Preferences for column density. Bulk actions for expert users.	Add: Ctrl+K command palette for quick navigation, recent items list.
H8	Aesthetic and Minimalist Design	4	Cloudscape's design system enforces visual discipline. Hero bar shows 5 KPIs (not 15). Table columns are configurable (hide low-value columns). Sparklines replace full charts in table cells for densit	Minor: Command Center dimension cards could be trimmed from 5 to top 3 on first load, with "Show all" toggle.
H9	Help Users Recognize, Diagnose, and Recover from Errors	4	Error Flashbar messages include: what failed, why (plain language), and a retry CTA. Integration errors show "Test Connection" button. Discovery errors show "Retry Scan" option.	Add: error codes with documentation links for IT admins troubleshooting integration failures.
H10	Help and Documentation	4	Contextual help icons (?) on complex elements (maturity scoring methodology, cost estimation approach). Onboarding tour for first-time users. Empty states include documentation links.	Add: searchable help panel (Cloudscape Drawer) with FAQ and quick-start guides.

Heuristic Score: 10/10 pass (≥ 3)

3C. Interaction State Matrix

Element	Default	Hover	Pressed/Active	Focus (keyboard)	Disabled
Primary button ("Apply Policy")	\$color-background-button-primary-default, white text, 2px border-radius, cursor:default	\$color-background-button-primary-hover, white text, cursor:pointer	\$color-background-button-primary-active, white text, scale(0.98)	\$color-background-button-primary-default + 2px \$color-border-control-focus outline offset 2px, cursor:default	\$color-background-button-primary-disabled, \$color-text-button-primary-disabled, cursor:not-allowed, opacity:1
Table row	\$color-background-container-content, \$color-text-body-default	\$color-background-dropdown-item-hover, cursor:pointer	\$color-background-item-selected, \$color-border-item-selected left border, \$color-text-body-default	\$color-background-container-content + 2px \$color-border-control-focus outline, cursor:default	\$color-text-status-inactive, cursor:not-allowed
Filter chip (Property Filter token)	\$color-background-badge-icon, \$color-text-body-default, pill shape	\$color-background-dropdown-item-hover, cursor:pointer	\$color-background-item-selected, border highlight	2px \$color-border-control-focus outline	\$color-background-input-disabled, \$color-text-status-inactive
Navigation link (sidebar)	\$color-text-body-default, no background	\$color-background-dropdown-item-hover	\$color-background-item-selected, \$color-border-item-selected left bar, \$color-text-link-primary-default	2px \$color-border-control-focus outline	N/A (nav items are never disabled, hidden instead)

Element	Default	Hover	Pressed/Active	Focus (keyboard)	Disabled
Split panel toggle (row click)	N/A (triggered by table row click)	See table row hover state	Split panel slides in, row gets selected state	Via table row focus → Enter triggers panel	N/A
Tab	\$color-text-body-default, transparent background, bottom border:none	\$color-text-link-primary-hover, cursor:pointer	\$color-text-link-primary-default, 2px bottom border \$color-border-tab-active	2px \$color-border-control-focus outline	\$color-text-status-inactive, cursor:not-allowed
Date range picker trigger	\$color-background-input-default, \$color-border-control-default, \$color-text-body-default	\$color-border-control-hover	\$color-border-control-focus, dropdown opens	\$color-border-control-focus 2px outline	\$color-background-input-disabled, \$color-text-status-inactive
Icon button (help ?, close X)	\$color-text-interactive-default, transparent background	\$color-background-dropdown-item-hover, \$color-text-interactive-hover	\$color-text-interactive-active, scale(0.95)	2px \$color-border-control-focus outline	\$color-text-status-inactive, cursor:not-allowed
Toggle (dark mode)	Track: \$color-background-toggle-default; thumb: white; label: \$color-text-body-default	Track: \$color-background-toggle-hover	Track: \$color-background-toggle-checked-default; thumb slides right	2px \$color-border-control-focus outline around track	Track: \$color-background-toggle-disabled; cursor:not-allowed
Alert dismiss button (Flashbar)	\$color-text-interactive-default, transparent	\$color-background-dropdown-item-hover	\$color-text-interactive-active, flash removed with fade-out	2px \$color-border-control-focus outline	N/A (dismiss always available)

3D. Keyboard Navigation Spec

Tab Order (Command Center page):

- Skip to main content link → main content area
- Side navigation items (7 sections) → vertical arrow key navigation within
- Date range picker → opens on Enter/Space
- Hero KPI cards (5) → arrow keys between cards, Enter to drill into section
- Flashbar alert items → Tab between alerts, Enter to view detail, Delete/Backspace to dismiss
- Dimension summary cards → Tab between cards, Enter to navigate to section
- Recent activity table → Tab to table, then arrow keys for row navigation
- Split panel (if open) → Tab cycles through panel tabs, panel content, close button

Keyboard Shortcuts:

Shortcut	Action	Scope	Conflict Check
/ or Ctrl+K	Focus global search / property filter	Global	No conflict — standard web pattern
Esc	Close split panel / modal / drawer	Contextual	Standard
Enter	Activate focused element / open split panel for focused row	Universal	Standard
Arrow Up/Down	Navigate table rows / navigation items	Within component	Standard
Arrow Left/Right	Navigate tabs / KPI cards	Within component	Standard
Ctrl+Enter	Submit / apply filters	Within filter	No conflict
? (Shift+)	Open help drawer	Global	No conflict with browser
g then c	Go to Command Center	Global (vim-style)	No conflict
g then i	Go to AI Inventory	Global (vim-style)	No conflict
g then g	Go to Governance	Global (vim-style)	No conflict

Focus Management Rules:

- When modal opens: focus moves to first focusable element (usually modal title or first form field). Modal traps focus — Tab cycles within modal only.

- When split panel opens: focus moves to panel heading. Split panel does NOT trap focus — Tab can leave panel back to main content.
- When split panel closes (via X or Esc): focus returns to the table row that triggered it.
- When Flashbar alert is dismissed: focus moves to next alert item, or to main content if no alerts remain.
- On page load / route change: announce page title via aria-live="polite", focus moves to .
- Skip links: "Skip to main content" (bypasses sidebar nav), "Skip to results" (bypasses KPI bar and alerts, lands on primary table).

3E. Anti-Pattern Validation

#	Anti-Pattern	Status	Notes
1	Mystery meat navigation (icons without labels)	Absent	Side navigation uses text labels for all 7 sections. Icons accompany labels but never replace them. Collapsed sidebar (M breakpoint) uses icon+tooltip, not icon alone.
2	Infinite scroll without progress indicator	Absent	All tables use pagination (Cloudscape Table pagination). Page count and total items displayed.
3	Hamburger menu hiding primary navigation	Absent	Sidebar nav is always visible on L/XL breakpoints. On S/XS, collapsed but accessible via hamburger — acceptable for mobile as standard responsive pattern, with all items immediately visible on expand.
4	Modal on top of modal (nested modals)	Absent	Design enforces single-modal rule. Policy action modal is the only modal surface; any secondary choices use inline expandable sections within the modal, not nested modals.
5	Confirm-shaming (guilt-trip copy on cancel buttons)	Absent	Cancel buttons use neutral "Cancel" label. No "No, I don't want to govern my AI" type copy.
6	Pagination that loses scroll position	Absent	Cloudscape Table pagination preserves scroll position. Page navigation does not reset scroll.
7	Auto-playing content without user initiation	Absent	No auto-playing animations, videos, or carousels. Sparklines render statically. Data refreshes on user action or configurable auto-refresh interval (default: manual).
8	Ambiguous destructive actions (delete without confirmation)	Absent	All destructive actions (block tool, delete policy, remove integration) require confirmation modal with impact summary.
9	Truncated content with no way to see full text	Absent	Long tool names truncate with ellipsis and show full name on hover tooltip. Table cells with truncated content show full text in split panel detail.
10	Ghost buttons for primary actions (low-contrast CTAs)	Absent	Primary actions use Cloudscape Primary Button (high contrast, filled). Secondary actions use Cloudscape Normal Button. No ghost buttons for important actions.

#	Anti-Pattern	Status	Notes
11	Data table without sortable columns	Absent	All table columns are sortable. Default sort: risk level (descending) for Security, cost (descending) for Cost Attribution, name (ascending) for AI Inventory.
12	Time-based UI without timezone clarity	Absent	All timestamps display in user's local timezone with "(UTC-5)" or equivalent suffix. Date range picker shows timezone. Governance activity log shows absolute timestamps with timezone.
13	Search with no empty-result guidance	Absent	Property Filter empty results show: "No tools match your filter" with suggestions: "Try removing filters" and "Check if the Discovery Engine has completed its scan."
14	Settings that require page reload to take effect	Absent	All settings apply immediately via API call with optimistic UI update. Policy changes propagate with a Flashbar confirmation ("Policy updated — changes will reach user devices within 24 hours").

3F. Data Visualization Decision Matrix

Data Point	Viz Chosen	Why	Alternative	Why Rejected
Total AI tools discovered (KPI)	Stat card with sparkline	Single KPI — stat card is the canonical pattern. Sparkline shows 30-day trend without dedicating a full chart.	Counter only	No trend context; a number alone doesn't show direction.
% AI tools governed (KPI)	Stat card with gauge arc	Percentage metric — gauge arc conveys progress toward 100% goal intuitively.	Stat card with sparkline	Gauge better communicates "progress toward target" than a line for percentage metrics.
Monthly AI spend (KPI)	Stat card with sparkline	Dollar amount over time — sparkline shows spend trend compactly.	Bar chart	Too much space for a single aggregate metric.
AI spend by vendor	Stacked bar chart (vertical, 5 vendors)	Categorical composition — stacked bar shows both total and per-vendor contribution. ≤5 vendors = vertical stacked bar works.	Pie chart	Pie chart difficult to compare similar-sized segments; stacked bar enables precise reading.
AI spend by team	Horizontal bar chart	Categorical comparison with 10+ teams — horizontal bar handles long team names and >7 categories.	Vertical bar	Team names truncate in vertical bar x-axis labels; horizontal is more readable for many categories.
Tool usage trend (per-tool)	Sparkline in table cell	Datadog density rule: sparklines-in-table-cells for correlated scanning. Each row gets a usage sparkline.	Separate line chart per tool	50+ separate charts is unreadable; sparklines in table enable pattern recognition at a glance.
Maturity distribution across teams	Horizontal stacked bar (5 levels)	Shows composition of maturity levels across teams — composition/proportion → stacked bar.	Donut chart	Donut works for one team; stacked bar compares across multiple teams simultaneously.
Risk alerts over time	Line chart (area fill)	Time-series trend for monitoring risk alert volume over time. Area fill emphasizes cumulative impact.	Bar chart	Bars obscure continuous daily trend in risk data.

Data Point	Viz Chosen	Why	Alternative	Why Rejected
Compliance rate trend	Line chart	Time-series trend — clean line shows whether compliance is improving.	Area chart	Area implies cumulative/volume which doesn't apply to a percentage rate.
Tool inventory (all tools)	Table with sort/filter	>5 columns of structured data per tool — table is the only appropriate pattern. Table supports sort, filter, pagination.	Card grid	Cards waste space for data-dense records; table is 3-5x more space-efficient for enterprise users.
Policy violations by type	Horizontal bar (ranked)	Showing rank/top-N violation types — horizontal bar with rank ordering.	Pie chart	Anti-pattern: pie with >5 slices.
Discovery source breakdown	Donut chart (3 segments)	Composition with exactly 3 sources (desktop detection, vendor API, manual) — donut appropriate for ≤5 slices.	Stacked bar	Overkill for 3 categories; donut is simpler and more intuitive for simple composition.

3G. Micro-Interaction Spec

Interaction	Trigger	Animation	Duration	Easing	Purpose
Split panel open	Click table row / detail link	Slide in from right (translateX(100%) → translateX(0))	200ms	ease-out	Reveal detail without losing list context
Split panel close	Click X / press Esc	Slide out to right (translateX(0) → translateX(100%))	150ms	ease-in	Quick exit, return focus to trigger row
Tab switch	Click tab	Fade content swap (opacity:0 → opacity:1)	100ms	ease-in	Fast, no layout shift
Modal open	Click action button	Fade in (opacity:0 → opacity:1) + scale(0.95 → 1) + backdrop fade	200ms	ease-out	Draw focus, indicate overlay
Modal close	Click X / Esc / Cancel / backdrop	Fade out (opacity:1 → opacity:0) + backdrop fade	150ms	ease-in	Quick exit, return focus
Filter apply	Click Apply / press Enter	Table body fades (opacity:0.5) during data load, then fades back (opacity:1)	150ms fade + data load time	ease-in-out	Signal results are updating
Flashbar notification appear	System event (discovery, policy change)	Slide in from top (translateY(-100%) → translateY(0))	300ms	cubic-bezier(0.34, 1.56, 0.64, 1)	Attract attention without blocking work
Flashbar notification dismiss	Click dismiss / auto-timeout	Fade out + collapse height (height → 0, opacity → 0)	200ms	ease-in	Smooth removal, no layout jump
Loading state	Data fetch begins	Skeleton screen replaces content (pulse animation on gray blocks)	Immediate swap, pulse: 1.5s loop	ease-in-out (pulse)	Preserve layout, indicate progress
KPI sparkline load	Data arrives	Draw-in from left (stroke-dashoffset animation)	400ms	ease-out	Visual interest, draw attention to trend
Sidebar nav hover	Mouse enter nav item	Background color fade to hover state	100ms	ease-in	Responsive feedback
Policy action confirm	Click "Apply" in modal	Button shows spinner → success checkmark → modal closes	200ms spinner, 300ms check, 150ms close	ease-out	Confirm action completed before modal disappears

Interaction	Trigger	Animation	Duration	Easing	Purpose
Page transition	Click sidebar nav item	Current content fades out, new content fades in	100ms out + 100ms in	ease-in-out	Smooth, no layout shift

Rules enforced:

- No animation > 400ms (except skeleton pulse loop which is a loading indicator, not a transition).
- All animations use CSS transitions, no JS animation libraries.
- Loading states use skeleton screens, never spinners (except inline button spinners for short async actions).
- No layout shift during transitions — elements do not reflow.
- All animations respect prefers-reduced-motion: reduce — if set, instant swaps replace all transitions.

3H. Responsive Breakpoint Specs

Component / Region	XL (>= 1600)	L (1200-1599)	M (992-1199)	S (688-991)	XS (< 688)
App Layout navigation	Side nav expanded (text + icons)	Side nav expanded (text + icons)	Side nav collapsed (icons only + tooltip on hover)	Side nav hidden, hamburger menu	Side nav hidden, hamburger menu
Split panel	Side position, 400px width	Side position, 360px width	Bottom position, 50% height	Bottom position, collapsed by default	Hidden; "View detail" navigates to full-page detail view
Hero KPI bar	5 cards in single row	5 cards in single row	3 cards in row 1, 2 in row 2	2 cards per row (3 rows)	1 card per row (stacked), compact variant
AI Inventory table	All 8 columns visible	All 8 columns visible	Hide: trend sparkline, discovery source (6 columns)	Hide: trend, source, risk level (5 columns)	Card view replaces table — one card per tool
Property filter	Full inline, all tokens visible	Full inline, all tokens visible	Compact — collapsed advanced options	Full-width stacked above table	Full-width stacked, simplified to search bar + dropdown
Cost attribution charts	Side-by-side (vendor bar + team bar)	Side-by-side	Stacked vertically	Stacked vertically, reduced labels	Sparklines only, table with key metrics
Maturity scorecard	5 cards in single row	5 cards in single row	3 + 2 row layout	2 + 2 + 1 row layout	Vertical stack, 1 card per row
Date range picker	Inline in header bar	Inline in header bar	Dropdown trigger in header	Dropdown trigger, full-width picker on open	Full-width picker, separate screen
Flashbar alerts	Full-width, multi-line allowed	Full-width, multi-line allowed	Full-width, single-line with "Show more"	Full-width, single-line truncated	Full-width, stacked vertically
Governance policy table	All 6 columns	All 6 columns	4 columns (hide: affected tools count, last modified)	3 columns (name, enforcement, status)	Card view

Enterprise note: Quick Desktop AI Control Tower is primarily used on L and XL breakpoints (desktop IT admin workstation). L is the optimization target. M is graceful degradation for laptop screens. S and XS provide usable access for on-the-go checks from tablets/phones but are not the primary use case.

3I. Accessibility Depth

ARIA Landmarks:

Landmark	Element	Role	Label
Main content		main	AI Control Tower
Navigation	(sidebar)	navigation	AI Control Tower navigation
Search	Property filter bar	search	Filter AI tools
Complementary	Split panel	complementary	Tool detail panel

Landmark	Element	Role	Label
Banner	Page header with breadcrumbs	banner	AI Control Tower header
Alert region	Flashbar container	alert (with aria-live="polite")	Notifications

Focus Management Rules:

- On page load: focus goes to (page title). Screen reader announces page name.
- On route change (sidebar nav click): announce new page title via aria-live="polite" region, move focus to new .
- On filter apply: announce result count ("[N] AI tools match your filter") via aria-live="polite".
- On error: announce error message via aria-live="assertive", move focus to Flashbar error.
- On split panel open: move focus to panel heading. On close: return focus to trigger row.
- On modal open: move focus to modal title. On close: return focus to trigger button.
- On Flashbar alert arrive: announce via aria-live="polite" ("New alert: [summary]"). Do not steal focus.

Screen Reader Announcements:

Event	Announcement	Priority
Data loading	"Loading AI tool inventory"	polite
Data loaded	"[N] AI tools loaded"	polite
Filter applied	"[N] tools match your filter"	polite
Error occurred	"Error: [message]. Retry available."	assertive
Tool selected	"[Tool name] selected, detail panel open"	polite
Policy applied	"Policy applied to [tool name]. [N] users affected."	polite
Shadow AI discovered	"New shadow AI detected: [tool name] used by [team]"	polite
Destructive action confirm	"Are you sure? This will block [tool name] for [N] users."	assertive

Color and Contrast:

- All text meets WCAG AA: 4.5:1 for normal text, 3:1 for large text (Cloudscape design tokens pre-validated).
- Status indicators never rely on color alone: risk levels use icon + text label (■ Critical, ■ Medium, ■ Low) rendered as Cloudscape StatusIndicator component with both color and text.
- Charts use Cloudscape color tokens AND pattern fills (diagonal lines, dots, crosshatch) for colorblind accessibility.
- Focus indicators use 2px solid outline with \$color-border-control-focus — visible on all backgrounds.

3J. Wayfinding & Navigation

Breadcrumb Spec:

- Breadcrumbs always visible in page header (Cloudscape Breadcrumb Group).
- Every segment except current page is a clickable link.
- On XS/S: truncate to last 2 segments with "..." overflow showing full path on tap.

Page Location Indicators:

- Active sidebar nav item highlighted with \$color-border-item-selected left bar and \$color-background-item-selected background.
- Page follows: [Current Page] - AI Control Tower - Quick Desktop
- URL reflects current state for deep-linking (see below).

Deep-Link Strategy:

State	URL Parameter	Example
Active section	/ai-control-tower/[section]	/ai-control-tower/inventory

State	URL Parameter	Example
Active tab	?tab=[tab-id]	?tab=shadow-ai
Applied filter	?filter=[encoded]	?filter=vendor%3DOpenAI
Selected tool	?selected=[tool-id]	?selected=chatgpt-plus-marketing
Time range	?start=[ts]&end;=[ts]	?start=2026-04-20&end;=2026-05-20
Split panel open	?detail=[tool-id]	?detail=jasper-ai-marketing
Maturity level focus	?level=[1-5]	?level=3

Every meaningful UI state is deep-linkable. Lisa shares URLs in Slack ("look at this shadow AI tool"), pastes in runbook documentation, and bookmarks her "Unapproved Tools" filter view. If a state can't be linked, it's a broken workflow.

Navigation Hierarchy:

- **Primary:** Cloudscape Side Navigation — 7 sections (persistent, collapsible)
- **Secondary:** Cloudscape Tabs within content area (e.g., AI Inventory tabs: All, Shadow AI, Approved, Blocked)
- **Tertiary:** Segmented control for view variants (e.g., Cost Attribution: By Vendor / By Team / By Tool)
- Never more than 3 navigation levels without breadcrumbs (enforced by breadcrumb spec above).

3K. First-Time User Experience

Progressive Onboarding Flow:

Stage	Trigger	What User Sees	Goal
1. Empty state	First visit, no integrations connected	Illustrated empty Command Center: "Welcome to AI Control Tower" hero with value prop ("Discover every AI tool your organization uses"), 3-step setup wizard CTA (Connect integrations → Enable desktop d	Connect first integration and enable discovery.
2. Discovery in progress	First integration connected, scan running	Command Center shows progress bar: "Scanning your organization... Found 12 tools so far." KPI hero bar populates incrementally. "Next step" banner: "Enable desktop detection to find shadow AI."	Build anticipation; show immediate value; guide to next step.
3. First results	Discovery complete, data populated	Optional tooltip walkthrough (5 steps): (1) Hero KPI bar, (2) Shadow AI alerts, (3) Click a tool for detail, (4) Apply a policy, (5) Check Cost Attribution. Dismissible, re-triggerable from Help menu.	Show the full governance workflow end-to-end.
4. Feature discovery	After 1 week of use	Subtle dot indicators on un-visited sidebar sections. "Did you know?" contextual tips on unused features (e.g., first visit to Maturity tab: "Your organization is at Level 2 — here's how to reach Leve	Expand usage breadth beyond Command Center and Inventory.
5. Power user	Regular usage (2+ weeks)	Keyboard shortcut hints appear on hover (e.g., hover over search bar shows "Tip: Press / to focus"). Saved view suggestions ("You filter by 'unapproved' often — save this view?"). Collection Preferenc	Increase efficiency for returning users.

Empty State Design (per section):

- **Command Center (no data):** Illustration of a radar scanning for AI tools. Headline: "Your AI radar is ready." Body: "Connect your first integration to start discovering AI tools across your organization." CTA: "Connect Integration" (primary), "Learn More" (link). Shows what the dashboard will look like when populated (ghost/skeleton version of KPIs).
- **AI Inventory (no tools):** Illustration of an empty magnifying glass. Headline: "No AI tools discovered yet." Body: "Once the Discovery Engine runs, this table will show every AI tool in use — including shadow AI you didn't know about." CTA: "Run

Discovery Scan" (primary), "Import tools manually" (secondary).

- **Governance (no policies):** Illustration of a shield outline. Headline: "No governance policies configured." Body: "Set your first policy to start governing AI usage. We recommend starting with an approved tool list." CTA: "Create First Policy" (primary), "Use recommended defaults" (secondary).

- **Cost Attribution (no cost data):** Illustration of a calculator. Headline: "No cost data available yet." Body: "Connect vendor admin APIs to see cross-vendor AI spend. Supported: AWS Bedrock, M365 Copilot, OpenAI." CTA: "Connect Billing Source" (primary).

- **Integrations (none connected):** Card grid showing available connectors (AWS, M365, OpenAI) with "Connect" buttons and estimated setup time per connector.

Onboarding Dismissal:

- All onboarding elements (tooltips, banners, dot indicators) can be permanently dismissed.
- Dismissal state persists in user preferences (stored via Quick Desktop's user settings API).
- "Getting Started" link in the Help menu (Cloudscape Help Panel) lets users re-trigger the onboarding tour.

Enterprise FTUX approach: Emphasis on competence, not delight. Configuration options visible from step 1. Documentation links prominent. Assume Lisa is skilled but unfamiliar with THIS product's specific workflow.

Product Navigation Map

#	Page/Section	Nav Label	Nav Icon	Proto v1 State	Eng v1 State	Competitor Precedent
1	Command Center	Command Center	Dashboard icon	Full — KPI hero bar, alert zone, dimension cards, recent activity	Full	ServiceNow: Dashboard hub; Datadog: Overview dashboard
2	AI Tool Inventory	AI Inventory	Search/magnifying glass icon	Full — table with split-panel detail, shadow AI tab, approved/blocked tabs	Full	ServiceNow: Discover pillar; Datadog: Traces list
3	Governance Policies	Governance	Shield icon	Full — policy table, create/edit policy, nudge configuration, compliance rate	Full	ServiceNow: Govern pillar
4	Security & Risk	Security	Lock icon	Full — risk alerts table, data sensitivity heatmap, threat detail panel	Full	ServiceNow: Secure pillar; Splunk: AI Defense
5	Cost Attribution	Cost Attribution	Dollar sign icon	Full — vendor breakdown chart, team breakdown chart, per-tool cost table	Full	ServiceNow: Measure pillar (cost); Datadog: per-token cost
6	Adoption & Maturity	Adoption	Chart-trending-up icon	Full — 5-level maturity scorecard, team comparison, adoption trend charts	Full	ServiceNow: Measure pillar (adoption). No competitor has maturity scoring.
7	Admin Console	Admin	Gear icon	Full — integration management, notification settings, team/user management, data retention config	Full	ServiceNow: Admin; Datadog: Integrations settings

#	Page/Section	Nav Label	Nav Icon	Proto v1 State	Eng v1 State	Competitor Precedent
8	Integration Marketplace	Integrations (sub-nav under Admin)	Puzzle piece icon	Placeholder — connector catalog cards with "Connect" for top 3, "Coming Soon" for 10+ others with logos and descriptions	Eng v1: top 3 connectors only	ServiceNow: 30+ integrations marketplace
9	Compliance Reports	Reports (sub-nav under Governance)	Document icon	Placeholder — report template previews with "Generate" button (mock export), sample report PDF preview	Not in Eng v1	ServiceNow: Compliance frameworks
10	Enforcement Rules	Enforcement (sub-nav under Governance)	Shield-check icon	Placeholder — enforcement rule builder with nudge/block/kill switch toggles, sample policy rule visualization	Eng v1: simplified rules only	ServiceNow: Kill switches, real-time enforcement
11	AI ROI Insights	ROI Insights (sub-nav under Adoption)	Lightbulb icon	Placeholder — mock productivity correlation cards ("Teams using ChatGPT complete reports 40% faster"), sample data layout	Not in Eng v1	ServiceNow: Measure pillar (just launched)
12	Policy Templates	Templates (sub-nav under Governance)	Template icon	Placeholder — NIST AI RMF, EU AI Act, SOC 2 template cards with "Apply Template" buttons, template detail preview	Not in Eng v1	ServiceNow: 5 NIST/EU AI Act frameworks
13	Activity Audit Log	Audit Log (sub-nav under Admin)	List icon	Full — governance action log with timestamp, user, action, target, filterable	Full	ServiceNow: Activity log

Completeness check: 13 navigation sections (7 primary + 6 sub-navigation). ServiceNow's AI Control Tower has 5 pillars + dashboard + admin + integrations = ~8 primary sections. Datadog's LLM Observability has ~6 sections. Our 13 sections (7 primary in sidebar + 6 sub-nav items) exceeds both competitors' navigation surface. Proto v1 has 8 fully functional pages + 5 placeholder pages = 13 total navigable pages.

5-Minute Demo Script

Step	Time	Page	Action	What to Say	Why It Matters
1	0:00	Command Center	Land on AI Control Tower	"This is your AI operations cockpit — everything about your organization's AI starts here."	Sets context, establishes the "cockpit, not control tower" positioning. Shows this is a product, not a feature.

Step	Time	Page	Action	What to Say	Why it Matters
2	0:20	Command Center	Point to hero KPI bar	"At a glance: 247 AI tools discovered, \$182K monthly spend across 5 vendors, 72% governed, maturity level 3. This is the answer to the CEO's question — no more 2-week investigations."	Demonstrates immediate executive-level value. Addresses JTBD #1 directly.
3	0:40	Command Center	Point to shadow AI alerts	"These are tools our Discovery Engine found that IT didn't know about — 4 new shadow AI tools in the last week. That's the 82% problem in action."	The "aha" moment. Shows discovery capability and the urgency it creates.
4	1:00	AI Inventory	Click nav → AI Inventory	"Let's drill into what AI we actually have. This is every AI tool in the organization — shadow and sanctioned — with usage, cost, and risk per tool."	Shows the full discovery breadth. Enterprise data density.
5	1:20	AI Inventory	Click "Shadow AI" tab	"I can filter to just shadow AI — 23 unapproved tools across 8 teams. Marketing has 6 alone."	Shows the governance gap in concrete terms.
6	1:40	AI Inventory	Click "Jasper AI" row → split panel opens	"Click any tool for the full picture: 12 users, \$2.4K/month estimated, no governance policy, medium data risk. And here's the action — I can approve, monitor, or block it right here."	Shows data depth per entity AND governance action in context. The split-panel pattern.
7	2:10	AI Inventory → Governance	Click "Apply Policy" → configure → confirm	"I'll approve Jasper with a data sensitivity nudge — users will get a gentle reminder when pasting sensitive data. Not a block; a guide."	Demonstrates the "cockpit" governance model — guide, don't just block. Key differentiator.
8	2:40	Governance	Navigate to Governance page	"Here's where I manage all governance policies. Approved tools, blocked tools, nudge rules, and compliance rate — 72% and climbing."	Shows governance as a managed system, not ad-hoc decisions.
9	3:00	Cost Attribution	Navigate to Cost Attribution	"Now the CFO's question: what are we spending? \$182K/month across AWS Bedrock, Copilot, OpenAI, and others. Broken down by team — Engineering is \$68K, Marketing is \$31K. Per-tool, per-team, per-vendor	Addresses JTBD #3. Cross-vendor cost visibility that no one else offers.

Step	Time	Page	Action	What to Say	Why It Matters
10	3:30	Adoption & Maturity	Navigate to Adoption & Maturity	"This is what nobody else shows you: your AI maturity score. We're at Level 3 — Adopting. Here's exactly what we need to do to reach Level 4: improve cost attribution coverage to 80% and enforce polic	Prescriptive maturity model — unique differentiator. Answers "what should we do next?"
11	4:00	Integration Marketplace	Navigate to Admin → Integrations	"We have 3 vendor integrations live today — Bedrock, Copilot, OpenAI. Coming soon: Anthropic, Google, Jasper, and 10+ more. One marketplace, all your AI vendors."	Shows the vision — cross-vendor is the moat. Placeholder pages demonstrate roadmap credibility.
12	4:30	Command Center	Return to Command Center	"Full circle — from that CEO question to governed AI in under 5 minutes. Discover, govern, measure, optimize. All from the same desktop agent your employees already use."	Ties the narrative together. Reinforces the "cockpit" positioning and the product loop.

Script quality test: Reading the "What to Say" column top-to-bottom tells a coherent product story: discover what AI exists → see the shadow AI problem → drill into specific tools → take governance action → manage policies → understand costs → measure maturity → see the integration vision → full circle. This is a product narrative (discover → understand → govern → measure → optimize), not a feature list.

Demo touches 7 different pages/sections (Command Center, AI Inventory, Governance, Cost Attribution, Adoption & Maturity, Integration Marketplace, back to Command Center), exceeding the minimum 4-page requirement.

Stickiness Design

Daily Habit Loop:

Trigger: Morning notification from Quick Desktop: "AI Control Tower: 2 new shadow AI tools detected overnight. 1 policy violation alert." (Leverages Quick Desktop's proactive notification system.)

Action: Open Command Center → scan hero KPIs and alert zone (10 seconds).

Variable Reward: Shadow AI discoveries create curiosity ("what did my teams adopt now?"). Governance % increasing creates progress satisfaction. New risk alerts create urgency.

Investment: Each policy configured, each tool approved/blocked increases switching cost. Governance history builds institutional memory that's expensive to replicate elsewhere.

Weekly Return Triggers:

- Monday morning digest: "Last week: 3 new tools discovered, compliance improved 2%, AI spend up \$4K. Your maturity score: unchanged at Level 3."
- Governance nudge effectiveness report: "Your nudges were accepted 73% of the time this week. Top accepted: data sensitivity nudge on ChatGPT."

Monthly Return Triggers:

- Executive summary auto-generated for Lisa's CEO report: pre-formatted AI governance summary exportable as PDF.
- Maturity advancement notification: "Congratulations — Engineering team reached Level 4 (Optimizing)."

Switching Cost Amplifiers:

- Governance policy library grows over time — recreating policies in another tool is days of work.
- Historical trend data (adoption curves, cost trends, compliance trajectory) creates irreplaceable longitudinal value.
- Integration configurations (API keys, team mappings, notification rules) are investment that doesn't transfer.

Error & Edge States

State	Design Treatment
Empty state (no data)	Illustrated empty state per section (see FTUX section 3K). Each includes: what page will show when populated, why it's empty, primary CTA to fix it, documentation link. Never a blank page or generic "
Error state (API failure)	Flashbar error at top of affected section: "Unable to load [data type]. [Plain language reason]. [Retry button]." Stale data stays visible with a "Data from [timestamp]" badge rather than replacing co
Loading state	Skeleton screen matching the final layout (gray pulsing blocks in place of KPIs, table rows, charts). Preserves spatial layout so no layout shift when data arrives. Never spinners (Cloudscape pattern)
Partial data (some integrations connected, others not)	Dashboard shows available data with "Connect [vendor] to see [data type]" inline prompts where data is missing. Cost Attribution shows connected vendors' data with "Connect OpenAI to see remaining \$38
Timeout (slow API response > 5s)	After 5s: skeleton screen adds "Taking longer than expected..." text. After 15s: timeout error with retry button. Background data fetch continues — if data arrives after timeout message, it replaces t
Permission denied	Inline message: "You don't have permission to [action]. Contact your Quick Desktop admin to request [specific permission]." Never a generic 403 page.
Discovery in progress	Animated progress indicator in Command Center: "Discovery scan in progress... [X] tools found so far. Estimated completion: [time]." Partial results display as they arrive.
Integration disconnected	Integration card shows "Disconnected" badge in red. Affected data in dashboard shows "[Vendor] data unavailable — reconnect integration" inline where vendor data would appear.
Stale data (> 24h since last refresh)	Yellow warning banner: "Data last refreshed [timestamp]. Some information may be outdated. [Refresh now] button." Does not block access to existing data.

Handoff Notes for Prototype Builder

Key interactions to implement (priority order):

Sidebar navigation between all 13 pages (7 primary + 6 sub-nav). Every page must be reachable.

Command Center hero KPI bar with 5 stat cards (static mock data, no sparkline animation needed in prototype).

AI Inventory table with sort (click column headers), filter (property filter with vendor/team/status tokens), pagination (10 per page), and row click → split panel.

Split panel with tabbed detail view (Overview, Users, Cost, Compliance tabs) showing realistic mock data per tool.

"Apply Policy" modal triggered from split panel — form with radio buttons (Approve/Monitor/Block) and confirmation.

Tab switching on AI Inventory (All / Shadow AI / Approved / Blocked) filtering the table.

Cost Attribution page with mock charts (stacked bar by vendor, horizontal bar by team).

Maturity scorecard with 5 level cards (current level highlighted).

Dark mode toggle in top-right corner.

Responsive behavior: Optimize for L breakpoint (1200-1599px). Split panel side position. Sidebar expanded.

Mock data requirements:

- 15-20 AI tools across 5 vendors (AWS Bedrock, Microsoft Copilot, OpenAI ChatGPT, Anthropic Claude, Jasper AI) and 6 teams (Engineering, Marketing, Sales, Finance, Legal, HR).
- Realistic names, costs, user counts, risk levels. Shadow AI tools (unapproved) should make up ~30% of inventory.
- 5-6 governance policies (varying enforcement levels).
- Cost data: total ~\$182K/month distributed across vendors and teams.

Placeholder pages: Integration Marketplace, Compliance Reports, Enforcement Rules, AI ROI Insights, Policy Templates — each needs sidebar nav entry, page header, realistic content preview with "Coming Soon" elements where appropriate. These are NOT blank pages.

Keyboard shortcuts: Implement / for search focus and Esc for split panel close at minimum.

Feedback to PRD

Assumptions Made Where PRD Was Ambiguous

Navigation structure: PRD mentions "a dedicated section within Quick Desktop" but doesn't specify navigation architecture. Design assumes a sidebar navigation within the AI Control Tower section (not tabs), with 7 primary sections + 6 sub-sections, matching enterprise console patterns.

Governance action flow: PRD describes "approve, monitor, or block" but doesn't specify the UX for applying policies. Design assumes an inline action model (act from the tool detail panel) rather than a separate policy management workflow. Both exist but inline is the primary flow.

Cost estimation UX: PRD mentions "estimates costs from usage patterns" for vendors without cost APIs. Design assumes these estimates are shown with a confidence indicator ("Estimated — based on usage patterns") to maintain CFO trust, as PRD FAQ Q16 implies but doesn't fully specify the UI treatment.

Maturity score visibility: PRD defines the 5-level model but doesn't specify whether it's per-org or per-team. Design implements both: org-level aggregate on Command Center, per-team breakdown on Adoption & Maturity page.

UX Constraints That May Require PRD Scope Changes

Proto v1 needs 13 pages to feel like a product. The PRD's scope table lists 14 capabilities but doesn't map to navigation pages. The design adds 5 placeholder pages (Integration Marketplace, Compliance Reports, Enforcement Rules, AI ROI Insights, Policy Templates) to Proto v1 scope. This doesn't change Eng v1 scope but extends the prototype surface. PRD Section "Scope Boundary" should be updated to reflect the 13-page navigation map.

Audit log is essential for enterprise governance. PRD doesn't explicitly mention an activity audit log, but enterprise governance buyers (CISO, compliance) require one. Design adds "Activity Audit Log" as an Eng v1 page under Admin. This is a new Eng v1 requirement.

Interaction Patterns That Imply New Requirements

Deep-linking requires URL state management. The design specifies that every UI state (filters, selected tool, active tab, time range) must be deep-linkable via URL parameters. This has engineering implications: the frontend must serialize/deserialize UI state from URLs, and the routing framework must support complex query parameters.

Saved views require user preferences storage. The design includes saved filter views (e.g., Lisa's "Unapproved Tools" view). This requires a user preferences API that may not exist in Quick Desktop today. PRD should add this as a dependency.

Keyboard shortcuts require a shortcut manager. The vim-style navigation shortcuts (g+c, g+i, g+g) and the / for search require a global keyboard shortcut system. This is a small but non-trivial engineering effort.